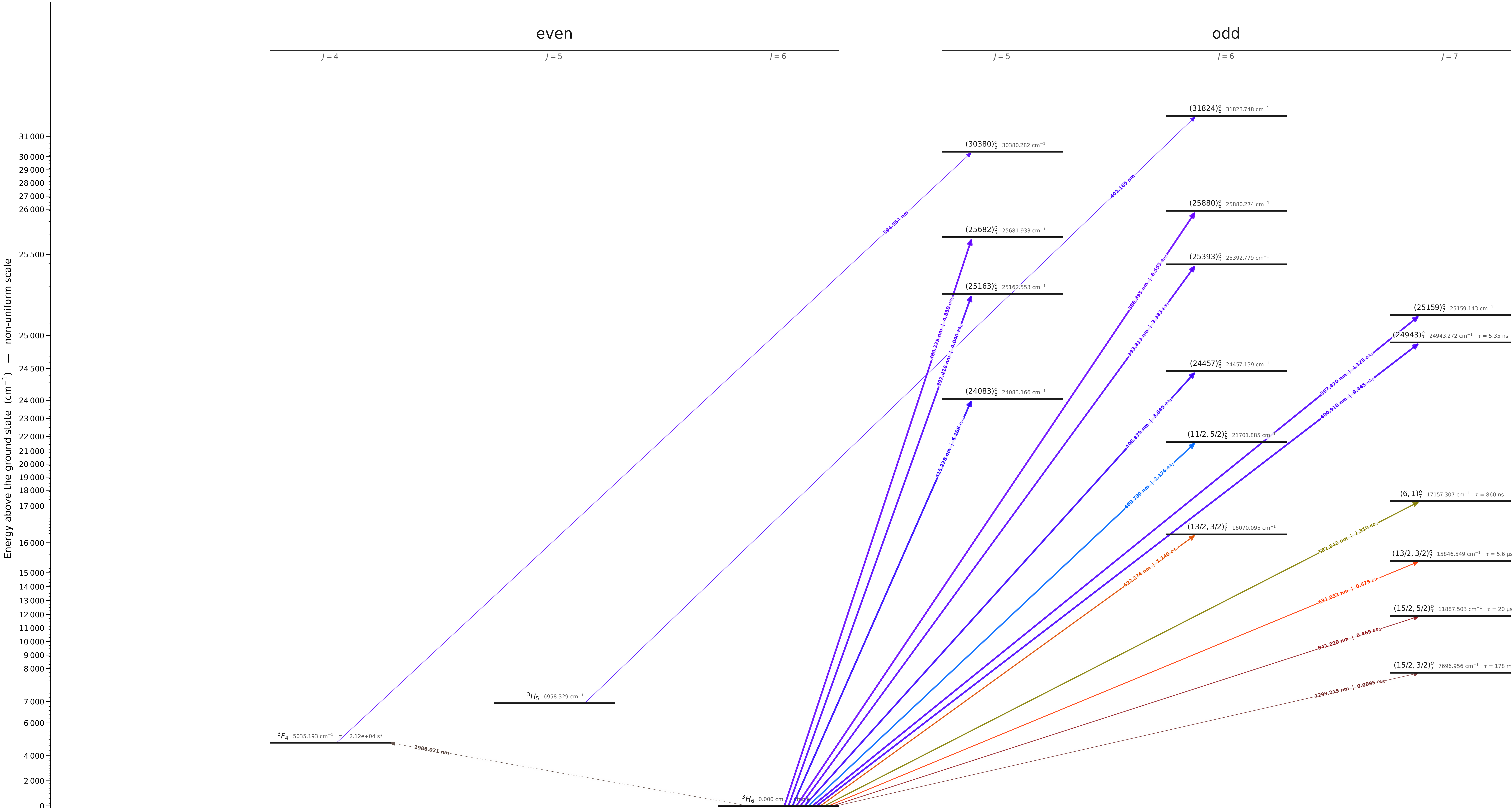
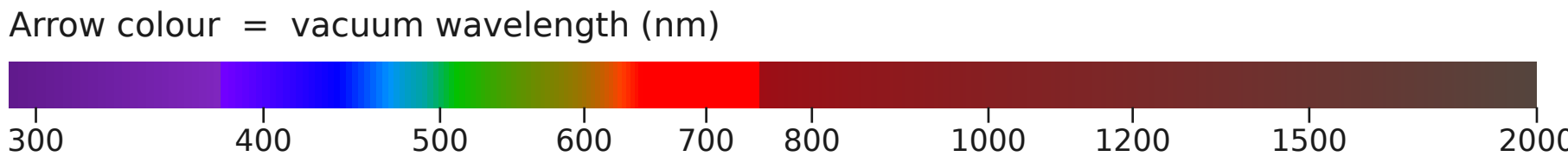


Erbium

Er (Er I). Strongest classified transitions below 2 μm (NIST): 19 levels, 17 lines.

Er⁺ ionisation limit 49262.00 cm⁻¹ = 6.10771 eV (202.996 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U\parallel er\parallel U'\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U\parallel er\parallel U'\rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ H ₆ – (24943) ^o _g	400.9097	400.7964	747.7805	9.445	1.870e+08	29.75	100	broad cooling line: Zeeman slower, transverse cooling, im...
³ H ₆ – (6, 1) ^o _g	582.8421	582.6805	514.3631	1.310	1.170e+06	0.1851	100	narrow-line MOT
³ H ₆ – (13/2, 3/2) ^o _g	631.0522	630.8778	475.0676	0.579	1.800e+05	0.02842	100	narrow line proposed for cooling / spin manipulation
³ H ₆ – (15/2, 5/2) ^o _g	841.2196	840.9884	356.3784	0.469	5.000e+04	0.007958	100	8 kHz narrow-line cooling / narrow-line MOT
³ H ₆ – (15/2, 3/2) ^o _g	1299.2149	1298.8596	230.7489	0.010	5.600e+00	8.941e-07	99.7	ultranarrow inner-shell
³ H ₆ – ³ F ₄	1986.0212	1985.4791	150.9513	–	4.700e-05	7.493e-12	99.8	proposed E2 optical clock transition within the 4f12 6s2...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
³ H ₆ – (24943) ^o _g	400.910	170-166	-1681(14)
³ H ₆ – (24943) ^o _g	400.910	168-166	-840(14)
³ H ₆ – (24943) ^o _g	400.910	167-166	-297(6)
³ H ₆ – (24943) ^o _g	400.910	164-166	849(17)
³ H ₆ – (6, 1) ^o _g	582.842	170-166	-1966(14)
³ H ₆ – (6, 1) ^o _g	582.842	168-166	-975(15)
³ H ₆ – (6, 1) ^o _g	582.842	167-166	-337(11)
³ H ₆ – (15/2, 3/2) ^o _g	1299.215	164-168	-2732.290(3)
³ H ₆ – (15/2, 3/2) ^o _g	1299.215	166-168	-1371.710(3)
³ H ₆ – (15/2, 3/2) ^o _g	1299.215	170-168	1414.920(5)

Shift = ν(first isotope) – ν(second isotope).

¹⁶⁷Er hyperfine structure (*I* = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
³ H ₆	-120.4870(10)	-4552.984(10)	F=5/2: +1141.09 F=7/2: +1495.46 F=9/2: +1773.69 F=11/2: +1842.74 F=13/2: +1540.01 F=15/2: +673.31 F=17/2: -979.08 F=19/2: -3668.47
³ F ₄	-121.900	516.000	F=1/2: +2386.64 F=3/2: +2138.64 F=5/2: +1738.45 F=7/2: +1205.84 F=9/2: +568.43 F=11/2: -138.21 F=13/2: -870.67 F=15/2: -1577.60
³ H ₅	-159.400	-4120.000	F=3/2: +1562.07 F=5/2: +1801.19 F=7/2: +1998.62 F=9/2: +2017.03 F=11/2: +1679.86 F=13/2: +771.28 F=15/2: -963.74 F=17/2: -3819.50
(15/2, 3/2) ^o _g	-125.851	-3046.052	F=7/2: +2385.74 F=9/2: +2368.13 F=11/2: +2215.11 F=13/2: +1847.77 F=15/2: +1172.87 F=17/2: +82.81 F=19/2: -1544.34 F=21/2: -3844.86
(15/2, 5/2) ^o _g	-126.560	-2969.000	F=7/2: +2434.38 F=9/2: +2399.70 F=11/2: +2229.14 F=13/2: +1845.79 F=15/2: +1158.77 F=17/2: +63.20 F=19/2: -1559.77 F=21/2: -3842.97
(6, 1) ^o _g	-172.70(7)	-4457.2(29)	F=7/2: +3170.27 F=9/2: +3196.05 F=11/2: +3035.13 F=13/2: +2572.06 F=15/2: +1670.40 F=17/2: +172.72 F=19/2: -2099.43 F=21/2: -5345.45
(24943) ^o _g	-100.1(3)	-3079(30)	F=7/2: +1652.40 F=9/2: +1756.61 F=11/2: +1751.05 F=13/2: +1555.96 F=15/2: +1077.10 F=17/2: +205.71 F=19/2: -1181.47 F=21/2: -3222.20

Measured A, B constants; sources in the data file.